TEMASEK JUNIOR COLLEGE, SINGAPORE
JC 2
Preliminary Examinations 2012
Higher 1

MATHEMATICS **8864/01**

Paper 1 **13 September 2012**

Additional Materials: Answer paper
List of Formulae (MF15) **3 hours**

READ THESE INSTRUCTIONS FIRST

Write your *Civics Group* and *Name* on all the work that you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **6** printed pages.

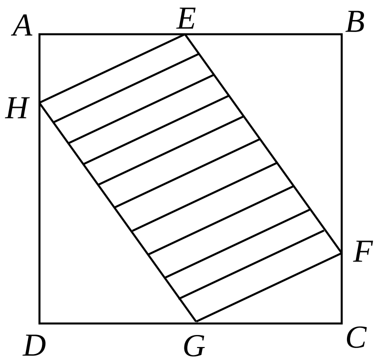
© TJC 2012

Pure Mathematics [35 marks]

1 Write $x^2+6kx+144$ in the form $(x+p)^2+q$, and obtain expressions for p and q in terms of k. [2]

Hence find the range of values of k such that $x^2+6kx+144$ is positive for all real values of x. [2]

2 A glass window in a new building is in the shape of a square *ABCD* with side 6 metres, as shown in the diagram. A tinted film *EFGH* is to be pasted on the glass to minimize the amount of radiation entering the building. The points *E*, *F*, *G* and *H* lie on *AB*, *BC*, *CD* and *DA* respectively such that $AH = CF = x$ metres and $AE = CG = 2x$ metres.



Find the largest area of the tinted film *EFGH* in exact form. [6]

3 **(a)** Differentiate $\dfrac{x^2-x}{x+1}$ with respect to x. [3]

(b) If $y=\dfrac{e^{t^2+mt}}{e^m}$, where m is a constant, find $\dfrac{dy}{dt}$. [2]

Hence find $\int (2t+m)\left(e^{t^2+mt}\right)dt$. [2]

4 Sketch the graph of $y=\cos x$ for $0 \le x \le 2\pi$. [1]

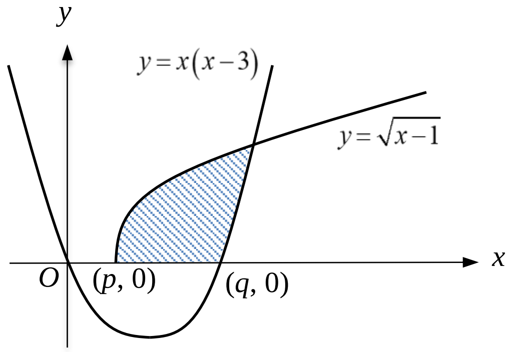
(a) Hence solve $\frac{1}{4}<\cos x<\frac{1}{2}$, expressing your answers correct to 3 decimal places. [2]

(b) **(i)** Sketch the quadratic curve $y=-k\left(x-\pi\right)^2+2$ on the same diagram, and state the coordinates of the maximum point. [2]

(ii) If α is a real root of the equation $-k\left(x-\pi\right)^2+2=\cos x$ for $0<\alpha<\pi$, state the other real root in terms of α. [1]

(iii) If $\alpha=\dfrac{\pi}{4}$, find the value of k. [2]

5 The curves C and D shown in the diagram below are described by the equations $y=\sqrt{x-1}$ and $y=x(x-3)$ respectively.



C cuts the x-axis at $(p, 0)$. D cuts the x-axis at the origin and $(q, 0)$.

State the values of p and q. [1]

State the coordinates of the point where the two curves intersect, correct to 5 significant figures. [1]

(a) Find the area of the region bounded by the two curves and the x-axis. [3]

(b) Calculate the value of the positive constant a such that the area of the region bounded by C, the x- and y-axes, and the line $y = a$ is 3 units^2. [3]

(c) Let d represent the vertical distance between the two curves at $x = t$, where $p \le t \le r$, and r is the x-coordinate of the point of intersection between the two curves. Using a numerical method, or otherwise, find the maximum value of d. [2]

Statistics [60 marks]

6 The perfect score of a game of bowling is 300. At Jackie's Bowl, it is known that the mean score is 128 and the standard deviation is 103.

Give a reason why a normal distribution with this mean and standard deviation, would not be a good approximation to the distribution of scores. [1]

A random sample of 50 games is taken. Calculate the probability that the mean score of this sample lies between 100 and 120. [3]

7 Twenty strips of cloth are placed in twenty boxes such that each box contains a strip. Each strip is cut into equal sections. Each section is then soaked in either red, blue or black dye, such that the number of red, blue and black sections in each box are in the ratio 2 : 3 : 2. A section is then randomly drawn from each box.

(a) Calculate the probability of drawing at least 4 but fewer than 7 blue sections. [3]

(b) Find the most likely number of blue sections that will be drawn. [1]

(c) If n strips are placed in n boxes instead where n is large, using a suitable approximation, find the maximum value of n such that the probability of drawing fewer than 6 blue sections exceeds 80%. [4]

8 **(a)** A and B are two events such that $\mathrm{P}\left(A \cap B'\right) = \dfrac{1}{5}$ and $\mathrm{P}\left(B \mid A\right) = \dfrac{1}{3}$.

(i) Find $\mathrm{P}\left(A \cap B\right)$. [2]

(ii) Given further that A and B are independent, find $\mathrm{P}\left(A \cup B\right)'$. [2]

(b) At a food kiosk serving confectionary for breakfast, 70% of customers choose white bread while the rest choose wholemeal bread. Out of those who choose white bread, 25% and 30% choose jam and butter spreads respectively, while out of those who choose wholemeal bread, the respective proportions are 10% and 20%. The rest chooses the chocolate spread. Each customer chooses one spread only.

(i) Find the probability that a customer chooses the chocolate spread. [2]

(ii) A customer does not choose chocolate spread. Find the probability that he chooses the jam spread. [3]

9 The weight of a chocolate bar produced by Holmes Enterprise is expected to be 40 grams. A random sample of 80 chocolate bars is taken and their weights, x grams, are recorded. The results are summarised by

$$\sum\left(x-40\right) = -17, \quad \sum\left(x-40\right)^2 = 167.$$

(i) Find the unbiased estimates of the population mean and variance. [2]

(ii) Assuming a normal distribution, test at 2% significance level whether the population mean weight of the chocolate bars is an underestimate. [4]

(iii) State, giving a reason, whether it is necessary to assume a normal distribution for the test to be valid. [2]

(iv) Find the set of values of sample mean for which the null hypothesis is rejected in favour of the alternative hypothesis at the 2% level of significance. [2]

10 **(a)** The random variable X is described by the distribution $\mathrm{N}\left(a-3, \sigma^2\right)$, where a is a constant. Given $\mathrm{P}\left(X<a\right)=3\mathrm{P}\left(X>a\right)$, find the value of σ. [3]

(b) A bakery sells only a particular kind of cupcake, and each cupcake consists of three parts: the cake (which forms the core), a thick icing sugar layer covering the top surface of the cake, and a paper cup which holds the cake. The masses of the cake and the icing sugar are assumed to follow normal distributions with means and standard deviations as shown in the table below, along with the production cost of each part.

	Mean (g)	Standard deviation (g)	Cost ($/g)
Cake	37	3.0	0.14
Icing sugar	9	1.3	0.10

Each paper cup has mass 1 g and costs $0.25.

(i) Find the mean and standard deviation of the mass of a randomly chosen cupcake. [2]
Hence, find the probability that a randomly chosen cupcake weighs within a gram of 48 g. [2]

(ii) Find the probability that the production cost of a randomly chosen cupcake exceeds $6.50. [3]

(iii) Find the probability that the difference in the production costs of two randomly chosen cupcakes exceeds $0.45. [4]

11 The Land Transport Authority (LTA) of Singapore announced on July 2012 that the next six-monthly Certificate of Entitlement (COE) will see a cut in the number of Category A (Cars below 1600cc & Taxis) COEs. There are usually 2 biddings per month.

(a) Gabriel, a self-directed student, who just learnt Statistics, wants to investigate the relationship between the COE quota and the Quota Premium (Quota Premium is the closing bid price) in Category A. He selects a sample of nine bidding results from the January 2008 to June 2012 by

- choosing at random one of the 12 biddings in first half of the year 2008,
- choosing every 12th bidding thereafter.

(i) What is this type of sampling method called? [1]

(ii) State one advantage and one disadvantage of the sampling method used in this context. [2]

(iii) Describe an alternative sampling method which would be better in this case. [2]

(b) The Quota Premium, y dollars, of 9 COE bidding results of different quota, x pieces, are given in the table below.

x	667	548	511	631	1148	1395	1839	2038	2053
y	56551	51000	44790	33089	20802	18020	4890	9501	11009

(i) Give a sketch of the scatter diagram for the data, as shown on your calculator. [1]

(ii) Find the product moment correlation coefficient and comment on its value in the context of the data. [2]

(iii) Find the equation of the regression line of y on x in the form $y = mx + c$, giving the values of m and c correct to 2 decimal places, and explain the meaning of m in the context of the data. [2]

(iv) Calculate an estimate of the quota premium for a quota of 390 pieces of Category A COE. Comment on the reliability of this estimation. [2]

(v) Explain why the regression line of x on y is not meaningful in the context of the data. [1]

(vi) Write down the values of mean of x, $\bar{x}$, and mean of y, $\bar{y}$, giving your answer to the nearest whole number. [1]

State the effect on the product moment correlation coefficient if an extra pair $(\bar{x}, \bar{y})$ is added to the set of data. [1]

END OF PAPER